Emanuele Zangrando

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Research Interests

My main area of interest is large-scale numerical optimization methods for deep learning, with a particular focus on Riemannian, bilevel, and non-euclidean methods. I also have a strong interest in implicit biases of optimization, mechanisms of feature learning in deep neural networks, and Hessian-free methods. On the side of applications, my main interest is the scientific computing acceleration of structured optimization problems.

Education

- **Gran Sasso Science Institute**, PhD program in Applied Mathematics. 11/2022-11/2026
  - Four-year PNRR scholarship on low-parametric and energy-efficient optimization methods for deep learning. During these years, I had the occasion to lead multiple research projects, which, as shown below, resulted in different publications in top-tier conferences.
  - Scientific Advisor: Prof. Dr. Francesco Tudisco, in collaboration with Dr. Gianluca Ceruti.
  - Pillar courses in: Numerical Analysis, Computational fluid dynamics, Statistical mechanics, and PDEs.
- **University of Padova**, MSc in Data Science (Mathematics of Data Science program). 01/2021 - 09/2022
  - Thesis title: "Dynamical low-rank training of neural networks", advised by Prof. Dr. Francesco Rinaldi and Prof. Dr. Francesco Tudisco.
  - Pillar courses: Numerical methods, high-dimensional statistics, optimization for data science, deep and machine learning.
- **University of Padova**, BSc in Mathematics. 10/2016 - 12/2020
  - Thesis title: "Modeling attention dynamics in social networks", advised by Prof. Dr. Marco Formentin.

Experience

Visiting period, University of Edinburgh, UK. September - January 2024

- Working on implicit biases of common optimization methods in deep learning and Riemannian methods for large-scale optimization.

Visiting period, University of Innsbruck, Austria. April - July 2024

- Investigating the efficiency of classical time-dependent variational principles coupled with neural network representations to approximate the solution of evolutionary PDEs for plasma physics problems.

Master thesis internship, Gran Sasso Science Institute, Italy. January - June 2022

- Working on variational methods to efficiently train low-rank compressed neural networks.

Publications

Neural-HSS: Hierarchical Semi-Separable Neural PDE Solver ICML 2026
P. Sittoni, ***E. Zangrando***, A.A. Casulli, N. Guglielmi, and F. Tudisco

GeoLoRA: Geometric integration for parameter efficient fine-tuning ICLR 2025
S. Schotthöfer ***E. Zangrando***, G. Ceruti, F. Tudisco, and J. Kusch

dEBORA: Efficient Bilevel Optimization-based low-Rank Adaptation ICLR 2025
E. Zangrando, S. Venturini, F. Rinaldi, and F. Tudisco

Geometry-aware training of factorized layers in tensor Tucker format NeurIPS 2024
E. Zangrando, S. Schotthöfer, G. Ceruti, J. Kusch, and F. Tudisco

Robust low-rank training via approximate orthonormal constraints NeurIPS 2023
D. Savostianova, ***E. Zangrando***, G. Ceruti, and F. Tudisco

HIJACK: Learning-based Strategies for Sound Classification Robustness to Adversarial Noise IEEE SMARTCOMP 2023

D. Sweet, ***E. Zangrando***, F. Meneghello

Low-rank lottery tickets: finding efficient low-rank neural networks via matrix differential equations ↗ — NeurIPS 2022

S. Schotthöfer, ***E. Zangrando***, J. Kusch, G. Ceruti, and F. Tudisco

Preprints

CRAdamW: A Cayley-Riemannian Extension of AdamW — NeurIPS 2026 under review

E. Zangrando, M. Sutti, F. Tudisco

Provable Emergence of Deep Neural Collapse and Low-Rank Bias in L^2-Regularized Nonlinear Networks ↗ — ArXiv 2026 (Under review)

E. Zangrando, P. Deidda, S. Brugiapaglia, N. Guglielmi, and F. Tudisco

Low-rank adversarial PGD attack ↗ — ArXiv 2024

D. Savostianova, ***E. Zangrando***, and F. Tudisco

Awards

ICML 2026 gold reviewer, NeurIPS 2025 top reviewer, NeurIPS 2023 scholar award, NeurIPS 2022 scholar award.

Professional Service

- Reviewing for ICML, ICLR, NeurIPS, AISTATS consistently since 2023. Reviewer of TMLR since 2026. Reviewed for AAAI, Scientific Reports, Neurocomputing, and SIAM Journal on Optimization.
- Organization of events: MS in SIAMLA (Paris, May 2024), MS in YAMC (Padua, September 2025).

Technologies

Languages: Python, R, Wolfram Mathematica, Matlab. Basic amateur knowledge of C. As Markup, good knowledge of LateX.

Technologies: Torch, JaX, Keras, Tensorflow and TensorFlow Probability, Bash, HPC, Git.

Scientific communication: talks, seminars and posters

- "Dynamical low-rank training of neural networks", SIAM Optimization conference, Edinburgh, June 2026.
- "Hierarchical structures in neural PDE solvers", Advanced Numerical Methods for Parametric PDEs, L'Aquila, February 2026.
- "Emergence and Stability of Deep Neural Collapse", Mathematics of data driven models, L'Aquila, January 2026.
- "Emergence and Stability of Deep Neural Collapse", YAMC, Padua, September 2025.
- "GeoLoRA: Geometric integration for parameter efficient fine-tuning", ICLR poster presentation, Singapore, April 2025.
- "dEBORA: Efficient bilevel optimization-based low-rank adaptation", ICLR poster presentation, Singapore, April 2025.
- "Deep learning for model order reduction in partial differential equations", GIMC-SIMAI young workshop, Naples, July 2024.
- "Online low-rank neural network compression", GIMC-SIMAI young workshop, Naples, July 2024.
- "Dynamical low-rank training of neural networks", SIAM Conference on Applied Linear Algebra (LA24), Paris, May 2024.
- "Robust low-rank training via approximate orthonormal constraints", NeurIPS poster presentation, New Orleans (USA), December 2023.
- "Dynamical low-rank training of neural networks", 2gg Algebra Lineare Numerica, L'Aquila, May 2023.
- "Low-rank lottery tickets: finding efficient low-rank neural networks via matrix differential equations", NeurIPS poster presentation, New Orleans (USA), December 2022.
- "Dynamical low-rank training of neural networks",Sciences - Computing - Data - Mathematics Seminar (SCDM), Karlsruhe Institute of Technology (Germany), November 2022.
- "Dynamical low-rank training of neural networks", GIMC-SIMAI young workshop, University of Pavia (Italy), September 2022.